TestBoard_ADC124 capture timing report

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What the four requested times mean

This board connects four sensor channels at once: each array's four external MUXes share address and enable controls. Each MUX output has an OPA365 buffer and a bias resistor; the ADC subsequently reads the buffers in sequence. A sensor therefore leaks through its bias resistor while earlier ADC inputs are sampled, even though its own sample has not started.

Requested quantity	Finding for this board	Confidence
Time connected to MUX output / op-amp	Physical EN/address connection until the bank next disconnects or changes address	Sequence established; numerical budget on pages 3-4
Time ADC takes the sample	Nominal 3 SCLK acquisition cycles, then 13 conversion cycles	Datasheet; actual hold edge includes clock/CS phase
Connection to retained sample	Turn-on guard + sensor settle + preceding words + retained repeat's acquisition	Route-dependent; array B waits for array A in shared MUX mode
Time disconnected	Next physical connection minus previous physical disconnection	Boundary routes can stay connected across sweep and USB work

Main result: settling is a bank-level wait, not the complete per-channel pre-sample leakage time. ADC input/MUX number, repeat, scan mode, sparse routing and sweep boundaries all change the four quantities. Current firmware has no ADC conversions during optional VMID visits. No new hardware measurements, firmware changes, or API functions were made for this report.

Hardware and settings

Item	Current repository value
Controller / converters	Teensy 4.1; two ADC124S101; one shared SPI bus
Array A / array B switches	Four TMUX1108 / four TMUX1308A; both banks supplied at 3.3 V
MUX outputs	MUX1-4 -> OPA365 buffers -> ADC IN1-4 respectively
Shared bias values A	MUX1-4: 470 kohm, 1 Mohm, 470 kohm, 1 Mohm
Shared bias values B	MUX1-4: 1 Mohm, 470 kohm, 1 Mohm, 470 kohm
Sensor addresses	0-5 on every MUX; address 6 only on MUX4; 25 routes per array
Hardware reference selections	Address 7 on all MUXes; address 6 on MUX1-3 also wired to VMID
ADC span / VMID	Fixed 3.3 V / 1.65 V; bias leakage toward VMID
Firmware and GUI defaults	ADC scan order, LPSPI, burst frames, 16 MHz requested, repeat 1, sensor settle 5 us, VMID off; VMID delay follows sensor delay

The local VMID-input capacitor is 150 pF. It is not permanently across the common sensor node; exact placement relative to the nearby 100 ohm resistor is unconfirmed. Do not invent that network's net order or use 150 pF as the sensor capacitance. [F1, F2, F4]

Capture process, step by step

Step	Current behavior	Time / count
1. Prepare run	Cache route groups; link the final command of each group to the next group's first ADC input; park both banks	Once per run; outside acquisition timestamps
2. Initialize selector	If unknown/different, one discarded conversion per active ADC selects its first input	At most 2 words at run start; no hidden prime per group
3. Begin acquisition	Record start; open one LPSPI session	Included in acquisition
4. Disconnect old address	Drive EN inactive	Additional programmed 0.200 us turn-off guard before address writes
5. Change address / reconnect	Cached address writes, then EN active	GPIO duration unmeasured; programmed 0.200 us guard after enable
6. Sensor settling	Wait once for participating bank(s)	S = muxsettle, 0-1000 us; skipped if address already enabled/unchanged
7. Read sensor group	Keep external address fixed, clock populated MUX inputs and repeats	16 clocks per conversion; only final repeat retained
8. Optional VMID visit	Disable, address 7, enable, guard, VMID wait	V = vmidsettle; follows S by default; no ADC clocks
9. Finish acquisition	Restore ordinary SPI state; record end	End includes all in-sweep switching, waits and reads
10. Encode and enqueue	Build frame; enqueue complete frame or discard if busy	Outside acquisition; 114 bytes for 50 routes
11. Stop / failure	Park both banks enabled at address 7	Outside ordinary steady-state sweep; unselected arrays disabled during runs

The 200 ns pre/post guards are firmware allowances, not measurements of switch turn-on/off. A zero analog wait retains both guards. Cached same-address selections can skip switching and settling entirely. [F1, F2, F3]

Physical execution order

ADC mode: A address 0, ADC IN1-4; A address 1, IN1-4; ... A address 6, IN4; then all B groups. Each bank has independent settling. A bank is not parked merely because the other ADC owns SPI.

MUX mode: enable the participating banks at address 0, share one settle wait, read A IN1-4, then B IN1-4; repeat addresses 1-5; at address 6 read A IN4 then B IN4. Both ADCs remain sequential on one bus. B sensors are connected while A is read. Sparse phases include only the banks present at that address. [F2]

Burst framing keeps CS low across all conversions in an ADC/address batch. Single framing pays CS setup/high allowances per word. The input selector written in each word chooses the next conversion. Prepared lookahead preserves it across VMID visits and sweep boundaries; there is no ADS7953 two-frame pipeline. [F2, F3]

ADC sampling window and per-channel equations

ADC124S101 acquisition nominally lasts 3 SCLK cycles; conversion occupies the next 13. With mode-0 SCLK low before CS, CS falling begins tracking and hold follows three physical falling clock edges. Burst words resume tracking at the preceding word boundary; clock gaps can stretch a window. Identify hold from actual CS/SCLK edges. [D1: pp. 5, 16-17]

Actual SCLK	One clock	Nominal acquisition	Conversion	16-clock word
8 MHz	0.125 us	0.375 us	1.625 us	2.000 us
16 MHz	0.0625 us	0.1875 us	0.8125 us	1.000 us

Let E be a bank's enable edge; t_on = E + d_on be physical connection; H the hold edge of the retained repeat; D the next disable edge; t_off = D + d_off. Then:

```
T_connected    = t_off - t_on
T_pre_sample   = H - t_on
T_ADC_track    = H - track_start
T_disconnected = next_t_on - t_off
```

For a new-address burst, let g_on=0.200 us, c_setup=0.100 us, W=16/f, a=3/f, r=repeat, and b the number of requested MUX inputs earlier in this ADC's group. With ideal uninterrupted words:

```
H - E = g_on + S + T_prior_ADC_batches + c_setup + (b*r + r-1)*W + a
T_pre_sample = (H - E) - d_on
```

`T_prior_ADC_batches` is zero in ADC mode and for A in full shared MUX mode. For B in a full address-0..5 MUX phase it includes A's 4r words, A's setup/high allowances, and software time between batches. It is not another settle wait. In single framing, earlier word costs also include their per-word setup/high overhead.

Numerical budget: S=5 us, repeat 1, 16 MHz, burst

The table measures enable-command to hold, with only configured guards and ideal clocks; GPIO/API/interrupt overhead is omitted. Both-bank enable skew is assumed zero. Subtract the corresponding switch turn-on delay to obtain physical connection-to-hold.

Scan / ADC	MUX1	MUX2	MUX3	MUX4
ADC mode, A or B	5.4875 us	6.4875 us	7.4875 us	8.4875 us
MUX mode, A	5.4875 us	6.4875 us	7.4875 us	8.4875 us
MUX mode, B	9.6875 us	10.6875 us	11.6875 us	12.6875 us

Each extra repeat adds `(b+1)*W` to retained hold time within that ADC's batch. B also waits for A's added repeats in MUX mode. At address 6 only MUX4 is read, so b=0 and B waits for one A word. These are calculated allowances, not measured per-channel times.

Connected dwell and sweep-boundary effects

All four MUXes in a bank connect the same address together, including sensor paths not requested for ADC readout. At addresses 0-5, the four sensors share approximately the same connected duration, but have different pre-sample durations. Sparse ADC selection does not independently disable unused MUX outputs. [F2, F3]

For a normal full four-input group followed immediately by an address switch, the enable-to-next-disable budget is:

```
ADC mode: g_on + S + (c_setup + 4*r*W + c_high)
MUX mode: g_on + S + 2*(c_setup + 4*r*W + c_high)
T_connected = enable_to_disable + d_off - d_on
```

At 16 MHz, S=5 us, r=1, c_setup=c_high=0.100 us, these budgets are **9.400 us in ADC mode** and **13.600 us in full shared MUX mode**. Add measured software gaps and bank skew. This applies to non-boundary groups; it is not a constant for every channel.

VMID on

After each phase, the participating banks disconnect sensors and select address 7 before waiting V us. This ends the sensor connection before the VMID wait. Thus V extends off-time/revisit time but does not directly extend that just-finished sensor's connected interval. It can still affect subsequent analog initial conditions.

VMID off

No end-of-sweep or ADC-ownership park is performed. In ADC mode, A stays at its last address throughout B's scan; B stays at its last address throughout the next A scan. With full routing this is address 6: MUX4's sensor stays connected much longer than ordinary sensors. MUX1-3 are connected to hardware VMID during this address-6 dwell.

In shared MUX mode, both banks remain at address 6 through encoding, USB enqueue and foreground work until the next sweep switches them to address 0. MUX4's sensor therefore includes that gap in its connected time. If a sparse bank uses only one external address with VMID off, its sensors can stay connected continuously across sweeps; unchanged selections skip the settle wait. Even unrouted sensors on that selected address can remain connected.

Disconnected time

For one contiguous connection per periodic visit, `T_disconnected = T_revisit - T_connected`. Compute the route's physical revisit and dwell; never subtract the ADC's 0.1875 us acquisition window instead of its full sensor-connected time. For continuous holding, off-time is zero and pre-sample age grows from the original connection.

As an arithmetic illustration only, a 100 us physical revisit with the page-3/4 ordinary ADC-group budget at S=1 us would have 5.400 us enable-to-disable dwell and about 94.600 us off-time, before switch delays and software overhead. This is not a measured per-route result from the 100 us G4 sweep.

Full-sweep timing budget for current firmware

Both full arrays contain 50 useful routes, 14 ADC/address batches, and 50r conversion words. VMID on adds **zero** conversion words. Burst framing uses 14 CS frames; single framing uses 50r. At repeat 1 this saves 36 pairs of configured setup/high delays, or 7.200 us, before engine overhead. [F1, F2, F3]

Configured contribution, full arrays / repeat 1	ADC order	Shared MUX order
16 MHz clocks, 50 words	50.000 us	50.000 us
Burst setup/high: 14 x 0.200 us	2.800 us	2.800 us
Sensor waits, if all addresses change	14S	7S
VMID waits when enabled	14V	7V
Switching guards, VMID off	14 x 0.400 = 5.600 us	7 x 0.400 = 2.800 us
Switching guards, VMID on	28 x 0.400 = 11.200 us	14 x 0.400 = 5.600 us

Shared mode overlaps switching guards and analog waits, not ADC words. ADC-mode guards are independent; full MUX phases pay the shared guard once after the last disable and once after the last enable. These budgets exclude register writes, sessions, batch handling, polling, interrupts, encoding and foreground work. They are not complete timing predictions and do not apply unchanged to sparse/same-address captures.

Compare with latest user-reported LPSPI rates

20 October 2026 observations: both full arrays, ADC order, repeat 1, no VMID ADC conversions. The G0/G4/G5 source gives rates but no raw edge traces, raw sweep timestamps or firmware image identity. For budgeting only, use current default 16 MHz and burst framing; that exact wire clock/framing remains an assumption for these observations. [M1]

Condition	Sensor S / VMID V	Reported rate per route	Reciprocal rate	Configured budget	Unassigned residual
G0: VMID off	5 us / none	6,736 Hz	148.456 us	128.400 us	20.056 us
G4: VMID on	1 us / 0 us	10,000 Hz	100.000 us	78.000 us	22.000 us
G5: VMID on	1 us / 5 us	5,880 Hz	170.068 us	148.000 us	22.068 us

The G5-G4 period increase is **70.068 us**, close to 14 x 5 us = **70.000 us** of added waits. This supports the ADC-order wait count. It does not isolate individual MUX switching, sample aperture or software phase costs. The reported rates are rounded observations; displayed reciprocal precision is arithmetic only.

At 8 MHz, the 50 repeat-1 words take 100 us rather than 50 us; at repeat 3 and 16 MHz they take 150 us. Analog wait counts stay unchanged for the same full routing. Engine overhead and actual divider selection still need their own measurements.

Historical measurements: keep firmware histories separate

The 14 October selector-lookahead run provides raw-backed production LPSPI evidence with 50 useful routes, repeat 1, burst, requested 16 MHz and 5 us sensor/VMID waits. Three repetitions had 1 s warm-up and 15 s measurement. Its historical VMID-on path clocked 56 extra VMID words, unlike today's wait-only path. [M2]

Historical mode	VMID	Sustained Hz per route, reported	SPI words per sweep
ADC	Off	6,748.7	50
ADC	On	3,332.8	106
MUX	Off	9,213.3	50
MUX	On	4,510.5	106

The historical off path supports the scale of G0's 6,736 Hz, but captures differ. Do not use the old VMID-on rates to calculate present wait-only dwell. Earlier timing tools also counted per-batch selector primes and/or five-word VMID scans; those counts are historical. The current prepared plan has neither in steady-state capture.

That historical series reported 7,091,330 acquired sweeps, 7,090,915 received frames and 415 accounted USB discards, with zero transfer/data/USB-write errors. It establishes transport and aggregate timing under the recorded configuration; it does not establish per-channel analog connection edges. The present review did not re-decode all those raw files.

Component timing relevant to physical boundaries

At 3.3 V, TMUX1108 enable-on is 14 ns typical / 25 ns maximum; enable-off is 7 ns typical / 12 ns maximum, with the datasheet's 200 ohm, 15 pF load conditions. These maxima cover the stated temperature ranges. Its off-source leakage is specified with voltage and temperature dependence. [D2: section 6.6]

TMUX1308A at 3.3 V specifies 26 ns typical / 40 ns maximum enable-on and 40 ns typical / 65 ns maximum enable-off at 25 C, using 10 kohm / 50 pF. Full-temperature maxima rise to 42 ns and 70 ns. Different test loads preclude treating the two typical numbers as measured bank skew. [D3: section 6.8]

For later off-time leakage calculations, the 3.3 V source-off leakage magnitudes are: TMUX1108, 1 pA typical / 50 pA maximum at 25 C, up to 500 pA over -40 to 125 C; TMUX1308A, 1 nA typical / 25 nA maximum at 25 C, up to 800 nA over -40 to 125 C. These use different datasheet terminal-voltage conditions and are not measured board currents. Preserve sign/voltage/temperature dependence rather than assuming constant discharge to ground. [D2, D3: section 6.6]

The 200 ns programmed guards exceed those tabulated switching delays, but full analog settling depends on sensor capacitance, resistance, buffer behavior and layout. The guards occur inside connection/leakage history: the post-enable guard is not all disconnected time. OPA365's typical 300 ns settling specification is for 0.01%, 5 V supply, gain 1 and a 4 V step; it is not a verified board settling time. [D4]

CS setup/high guards are each 100 ns in firmware. Actual first-clock phase, burst FIFO gaps, DMA byte gaps and inter-batch time must be measured. The converter has no returned channel tag; route correctness needs decoded DIN plus a known analog input, not response values alone.

Leakage interpretation and inputs for later functions

While connected, the bias resistor pulls toward VMID. Disconnected sensor leakage toward ground and MUX off-source leakage are distinct contributions; off leakage need not discharge in only one direction. A first-order sensor model is:

```
connected:    dV/dt = -(V-Vmid)/(R_bias*C_eff) - V/(R_sensor*C_eff) + I_on/C_eff
disconnected: dV/dt = -V/(R_sensor*C_sensor) + I_off/C_sensor
```

Use per-MUX resistor metadata, sensor capacitance/insulation, switching charge and buffer-input parasitics. ADC sampling capacitance is behind OPA365 and must not be directly added to the high-impedance sensor node. Connection charge sharing and the shared-node voltage left by previous sensors or VMID can change the initial voltage; a simple decay factor alone will not predict all ghosting.

Input for future functions	Present evidence	Missing / must be recorded
Route / MUX / array order	Prepared plan and payload routes	Exact effective configuration per run
Sensor and VMID delay	muxsettle / vmidsettle and follow-sensor state	Measured delay-call / GPIO overhead
Sample offsets	16-clock words; nominal 3-cycle acquisition	Real clock, first-CS phase, hold edges, FIFO/DMA gaps
Sensor dwell and off-time	EN/address sequence; shared-bank topology	Per-bank enable skew, switch delays, boundary cadence
Analog leakage parameters	Bias values and component datasheets	Actual sensor C, insulation R, assembly tolerance, temperature
Local VMID network	150 pF near VMID input; address-6 ties on MUX1-3	Confirm resistor/capacitor net connectivity before modeling
Transport effects	Nonblocking whole-frame discard policy	All-sweep cadence, including discarded frames

Focused measurement needed for exact per-channel timing

1. Record firmware image/source identity, protocol version, exact routes, scan order, burst/single framing, engine, repeat, S, V, VMID and effective clock request. Preserve address-6 wiring as hardware metadata.

2. Capture EN pins 0/32, address pins 1/2/3 and 29/30/31, SCLK13, CS9/24 and DIN11. Resolve 100/200 ns guards and actual 8/16 MHz clocks. Decode retained repeat and selector lookahead for each group.

3. Observe a known sensor input, MUX common/buffer input and buffer output with low-capacitance probes. Correlate physical make/break and settling with EN and hold edges; account for probe loading.

4. Test ADC/MUX modes, VMID off/on, V=0/5 us, S=1/5 us, repeats 1/3, sparse and single-address routes. Include MUX4 address 6, across-array idle time and sweep/USB boundaries. Report typical and tail distributions.

5. Fit phase overhead only from matching builds and settings. Validate per-channel leakage predictions using measured sensor C/R and a repeatable injected signal before implementing future firmware/API functions.

The future result should distinguish physical connection, ADC track start/end, retained hold, result-ready and disconnection; include the active bank's shared history and a continuous-hold state. No calculation functions are implemented at this stage.

Sources and review scope

Reviewed repository HEAD: `bc74949548091bb43aaffdfc37b169fce83f3a21`. Initial working tree was clean. The adjacent source manifest preserves reviewed-file SHA-256 hashes. No serial port was opened and no firmware uploaded. Current code is authoritative for operation counts; older reports remain evidence only of their recorded firmware/configurations.

Firmware and hardware sources

- **F1.** Hardware constants, `Arduino_Sketches/TestBoard_ADC124/include/ConfigurableParameters.h`: pin map, polarity, resistor values, delay/clock constants.

- **F2.** Acquisition application, `Arduino_Sketches/TestBoard_ADC124/src/main.cpp`: prepared groups/phases, selector lookahead, wait-only VMID, start/stop and timestamps. Read this for current behavior; older README text still mentions VMID conversions in places.

- **F3.** ADC/MUX driver and SPI controller, `Arduino_Sketches/TestBoard_ADC124/src/`; GUI profile, `config/boards/profiles/testboard_adc124.json`.

- **F4.** Board designer hardware clarifications, `docs/testing/results/testboard-adc124/mux_type_comparison_2026-10-15/README.md`: buffer isolation, 3.3 V MUX supply, local 150 pF, uncertain resistor/capacitor order, address-6 VMID ties.

Measured evidence

- **M1.** 20 October ghosting comparison notes, engine/rate CSV and rerun provenance, `docs/testing/results/testboard-adc124/ghosting_manual_2026-10-20/`. User-reported LPSPI rates, not new instrumented captures.

- **M2.** 14 October selector-lookahead LPSPI report and capture metrics, `docs/testing/results/testboard-adc124/selector_lookahead_lpspi_2026-10-14/`. Historical VMID conversion path differs from current code.

Primary component references, checked 21 October 2026

- **D1.** TI ADC124S101, SNAS283D, March 2013; pp. 5, 16-17: clock-cycle acquisition/conversion and next-input selection.

- **D2.** TI TMUX1108, SCDS388B, February 2024; section 6.6, 3.3 V timing and leakage.

- **D3.** TI TMUX1308A, SCDS477A, June 2025; sections 6.6 and 6.8, leakage and switching conditions.

- **D4.** TI OPA365, SBOS365G, May 2023; section 7.6, component settling conditions.

Interpretation of transmitted timing

Wire start/end cover acquisition, including in-sweep GPIO settling and VMID parking, but exclude USB enqueue. `avg_dt_us` is the rounded acquisition duration / route count, not the ADC track window or uniform channel spacing. A received-frame gap may contain discarded sweeps that still connected and sampled sensors. Sampling-period counters cover every completed sweep but do not expose per-channel edges. Unsigned microsecond timestamps wrap at 2^32 us (about 71.6 minutes).